

REVIEW

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Next generation building materials for energy efficiency and climate responsive design

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Abstract

The growing demand for sustainable construction, driven by climate change and urban expansion, highlights the urgent need for next generation building materials that enhance energy efficiency while supporting environmental goals. Traditional materials often fall short in addressing dynamic thermal loads, resource efficiency, and occupant comfort. This review investigates recent advancements in energy-efficient and climate responsive building materials, focusing on their insulation properties, thermal regulation, durability, and ecological impact. The objective is to assess the performance, benefits, and limitations of materials such as phase change materials (PCMs), aerogels, vacuum insulation panels, nanocomposites, and bio-based alternatives including straw, cork, and recycled steel. Based on findings from experimental, numerical, and analytical studies, these materials demonstrate significant potential to reduce energy consumption, improve indoor thermal comfort, and contribute to low-carbon construction practices. However, challenges such as high upfront costs, scalability issues, and lack of standardized performance metrics remain. The review recommends prioritizing life-cycle performance assessments, cost-effective production, and integration into building codes to support practical deployment. This work provides actionable insights for architects, engineers, and policymakers aiming to implement energy smart and climate-resilient design strategies through advanced material selection.

Article highlights

- This review explores the role of advanced energy materials in promoting sustainability in the construction sector.
- It categorizes energy materials based on their applications in insulation, energy harvesting, energy storage, and thermal regulation within buildings.
- Emphasis is placed on phase change materials (PCMs), aerogels, and nanocomposites for enhancing thermal efficiency and reducing energy consumption in built environments.
- Sustainable and low-carbon alternatives such as bio-based insulation materials, recycled composites are addressed.

Keywords Energy-efficient building materials, Phase change materials (PCMs), Bio-based insulation, Sustainable construction, Thermal insulation technologies, UN sustainable development goals (SDG 9)



1 Introduction

The detrimental effects of various developments on the ecosystem have given sustainability a lot of significance. The active construction that has accompanied the recent decade's tremendous growth has occasionally overlooked the impacts of climate change and living beings. Even though traditional heritage is a valuable resource for sustainable building techniques, the effects of advancements on it have not been taken into account. To satisfy the rapid urban growth, there is a need to create environmentally friendly building materials due to the ongoing exploitation of natural resources. With a focus on using natural resources effectively while also safeguarding global ecosystems, sustainable structures are constructed, built, and maintained [1]. Choosing the right construction materials promotes effective energy use [2, 3]. Researchers are searching for novel technologies and materials to use in future constructions that have advantages like energy conservation, resource, and water management, improved quality of indoor air, life cycle cost reduction, and durability in the rapidly changing building sector [4–7].

The research works published on energy-efficient building materials over the period 1976 to 2024 are illustrated in Fig. 1. An increasing trend of research interest in this field can be observed from this graph. This growth reflects the rising global concern about climate change and the urgent need for sustainable construction practices. Research acceleration after 2000 can be attributed to technological advances, environmental policies, and the adoption of green building codes.

It has been deeply analyzed from the Scopus database about the type of documents published in this emerging field and illustrated in Fig. 2. Among all publications, it was found highest 52% is for journal articles followed by conference papers which is 33%. This distribution suggests that academic and applied research in energy-efficient building materials is rapidly maturing, with journal articles indicating in-depth studies and peer-reviewed validation.

Energy-efficient building materials are materials used in the construction of buildings that are designed to reduce energy consumption and improve sustainability. These materials are specifically chosen to enhance the energy efficiency of a building by reducing

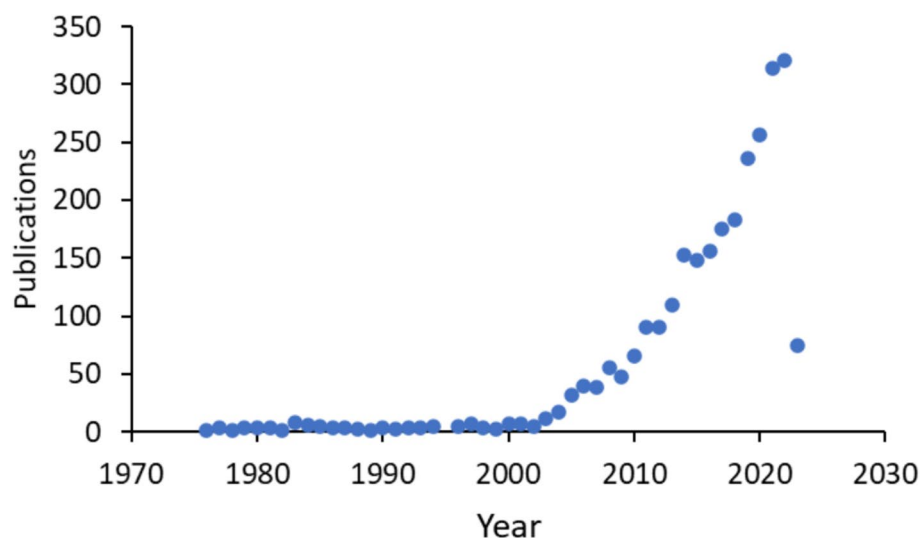


Fig. 1 Year-wise number of articles published related to energy-efficient building materials taken from the Scopus database

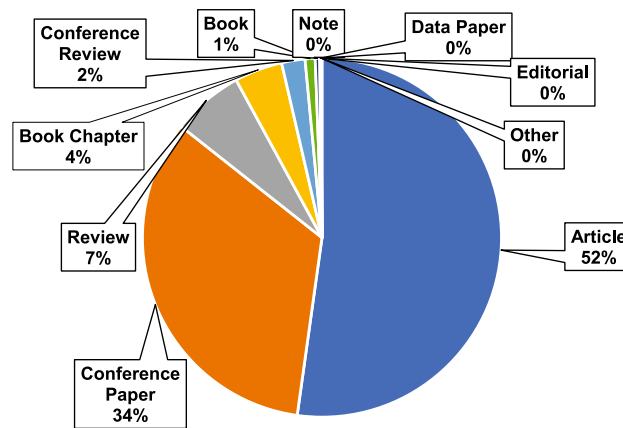


Fig. 2 Statistics of publications from Scopus database (keyword: energy efficient building materials)

heat transfer, preventing air leaks, and improving insulation [8]. The use of energy-efficient building materials is an essential component of sustainable design and construction [9]. It is crucial to reduce the environmental impact of buildings and achieve energy efficiency goals. Energy-efficient building materials can help reduce energy costs, improve indoor comfort, increase durability, and improve property value [10]. With an increasing focus on sustainability and reducing greenhouse gas emissions, the use of energy-efficient building materials is becoming more important than ever [11, 12]. By incorporating energy-efficient building materials into the design and construction of buildings, it is possible to create sustainable and environmentally friendly structures that are comfortable, durable, and cost-effective [13]. Several energy-efficient building materials can be used in construction to improve the energy efficiency of buildings. Some examples of energy-efficient building materials include insulated concrete forms (ICFs), structural insulated panels (SIPs), cool roofs, low-E windows, recycled insulation, green roofs, and aerated autoclaved concrete (AAC) [14]. These materials are designed to reduce heat loss or gain, minimize air leaks, reflect sunlight, and absorb rainwater, among other benefits.

From the literature available it is evident that there is a shortage of research that takes into account different energy-efficient materials, their application, and benefits. In the current paper significant benefits of energy-efficient building materials are described. The methodology adopted for this review includes a systematic literature review of peer-reviewed articles, conference proceedings, and industry reports published between 2015 and 2024, focusing on materials with demonstrated energy-saving potential. To enhance clarity and organization, energy-efficient building materials have been grouped into four primary categories based on their function and characteristics. This structured classification aids in better understanding their applications, advantages, and limitations. The energy performance of these materials when integrated into building systems were examined by evaluating their thermal conductivity, insulation properties, and overall contribution to building energy efficiency.

2 Research methodology

This review study employed a structured and comprehensive approach to identify, analyze, and synthesize recent developments in energy-efficient building materials. The methodology involved an extensive literature review focusing on peer-reviewed journal articles, conference papers, industry reports, and technical documentation published

between 2015 and 2025. Major academic databases such as Scopus, Science Direct, and Web of Science were searched using a combination of keywords including “energy-efficient building materials,” “thermal insulation,” “low-emissivity windows,” “phase change materials,” “sustainable construction,” and “advanced insulation technologies.”

The materials included in this review were selected based on a set of inclusion criteria relevance to current energy performance goals in buildings, demonstrated application in construction practices, and quantifiable benefits in terms of thermal performance, environmental impact, and sustainability. Publications that lacked experimental, analytical, or numerical data, or those focused on outdated or region-specific legacy materials were excluded. Emphasis was placed on studies that provided insight into real-world performance, durability, lifecycle considerations, and innovations supporting the UN Sustainable Development Goals.

The materials presented were chosen based on their prominence in contemporary research, documented effectiveness in reducing energy consumption, and growing adoption within the construction sector. These include Insulating Concrete Forms (ICFs), PIR Foam Energy Bricks, Straw and Stucco, Compressed Soil Blocks, Structural Insulated Panels (SIPs), Vacuum Insulated Panels (VIPs), Plant-Based Polyurethane Insulation, Plastic Composite Wood Substitutes, Low Emissivity Windows, and Recycled Steel. Each of these materials was evaluated based on parameters such as thermal conductivity, resistance to environmental degradation, ease of integration into existing systems, cost implications, and lifecycle environmental benefits.

Data collected from the literature were analyzed thematically and quantitatively, where possible. The comparative assessment of the selected materials is presented in tabular form to highlight key benefits, limitations, and implementation challenges. Numerical and experimental studies cited within the manuscript further validate the performance metrics of these materials. This methodological framework ensures transparency, replicability, and a comprehensive understanding of how the conclusions in this review were drawn.

3 Significant benefits of energy-efficient building materials

Energy-efficient building materials offer a wide array of benefits that contribute to the creation of more sustainable, comfortable, and cost-effective structures. These materials play a crucial role in addressing the global challenges of climate change and resource depletion while simultaneously enhancing the quality of life for building occupants [15].

One of the most significant advantages of energy-efficient building materials is their ability to substantially reduce energy consumption in buildings. By providing superior insulation and minimizing heat transfer, these materials significantly decrease the demand for heating and cooling. This reduction in energy usage translates directly into lower utility bills for building owners and occupants. Moreover, the decreased energy consumption leads to a marked reduction in greenhouse gas emissions, contributing to global efforts to mitigate climate change [16].

The implementation of energy-efficient building materials also results in notably improved indoor comfort. These materials effectively reduce drafts and temperature fluctuations, creating a more stable and pleasant indoor environment. This enhanced comfort level has far-reaching implications beyond mere convenience. Studies have shown that improved indoor environmental quality can positively impact occupant

health, well-being, and productivity. In residential settings, this can lead to better sleep quality and overall life satisfaction, while in commercial spaces, it can contribute to increased worker productivity [17].

Many energy-efficient building materials, such as insulated concrete forms and structural insulated panels, offer the added benefit of increased durability. These materials are often more resistant to weathering, moisture, and pests compared to traditional building materials. This enhanced durability translates into buildings that maintain their structural integrity and energy efficiency over longer periods, reducing the need for frequent repairs and replacements. Consequently, this leads to lower maintenance costs over the building's lifecycle and contributes to the overall sustainability of the structure by reducing the demand for replacement materials and the associated environmental impacts of construction and renovation activities.

The environmental benefits of energy-efficient building materials extend beyond energy savings and reduced greenhouse gas emissions. By decreasing the demand for energy, these materials indirectly reduce the need for non-renewable resources such as fossil fuels used in energy production. Furthermore, many energy-efficient materials are produced using sustainable practices or recycled content, further minimizing their environmental footprint. Some advanced materials even have the capacity to actively improve air quality by absorbing pollutants or reducing the formation of harmful compounds indoors [18].

From an economic perspective, the use of energy-efficient building materials can significantly enhance property values. As awareness of environmental issues grows and energy costs continue to rise, buildings incorporating these materials become increasingly attractive to potential buyers and renters. Properties with high energy efficiency ratings often command premium prices in the real estate market and tend to sell or rent faster than their less efficient counterparts. This increased market value provides an additional incentive for property owners and developers to invest in energy-efficient materials, creating a positive feedback loop that encourages wider adoption of sustainable building practices. The benefits of energy-efficient building materials are multifaceted, addressing environmental, economic, and social aspects of sustainability. As technology advances and these materials become more widely available and cost-effective, their role in creating a more sustainable built environment is likely to become even more pronounced, driving the transformation towards a low-carbon, resource-efficient future.

4 Global energy scenario

The global energy scenario for energy-efficient building materials is growing rapidly, driven by the increasing demand for sustainable and energy-efficient buildings. According to a report by the International Energy Agency (IEA), buildings are responsible for approximately 40% of the global energy consumption and 30% of greenhouse gas emissions [19]. This makes the use of energy-efficient building materials crucial to achieving global energy and climate goals. The market for energy-efficient building materials is expected to grow significantly in the coming years. According to a report by Markets and Markets, the global market for energy-efficient building materials is expected to reach \$101.3 billion by 2026, growing at a compound annual growth rate (CAGR) of 9.3% from 2021 to 2026.

The use of energy-efficient building materials is being encouraged by governments around the world through building codes, standards, and incentives. For example, in the United States, the Energy Policy Act of 2005 requires federal buildings to meet energy efficiency standards, and many states and municipalities have adopted similar codes and standards. In the European Union, the Energy Performance of Buildings Directive requires member states to establish minimum energy performance requirements for buildings. The use of energy-efficient building materials is also being driven by the growing awareness of sustainability and climate change. Consumers are increasingly seeking environmentally friendly and energy-efficient homes and buildings, and this is driving demand for energy-efficient building materials.

The global energy scenario for energy-efficient building materials is strong, and the market is expected to continue to grow as the demand for sustainable and energy-efficient buildings increases.

5 Recent energy-efficient building materials

Recent energy-efficient building materials are designed to reduce energy consumption and enhance sustainability. Materials like aerogel and phase change materials help maintain indoor temperatures by improving insulation. Low-emissivity (Low-E) glass allows natural light while blocking heat, making it ideal for windows. Different types of Energy-efficient building materials are shown in Fig. 3 and their advantages are discussed in Table 1. Figure 3 provides a visual classification that helps understand the broad categories based on material function such as insulation, thermal regulation, and structural application. This schematic representation supports a systems-level view of how various materials contribute to the energy performance of buildings.

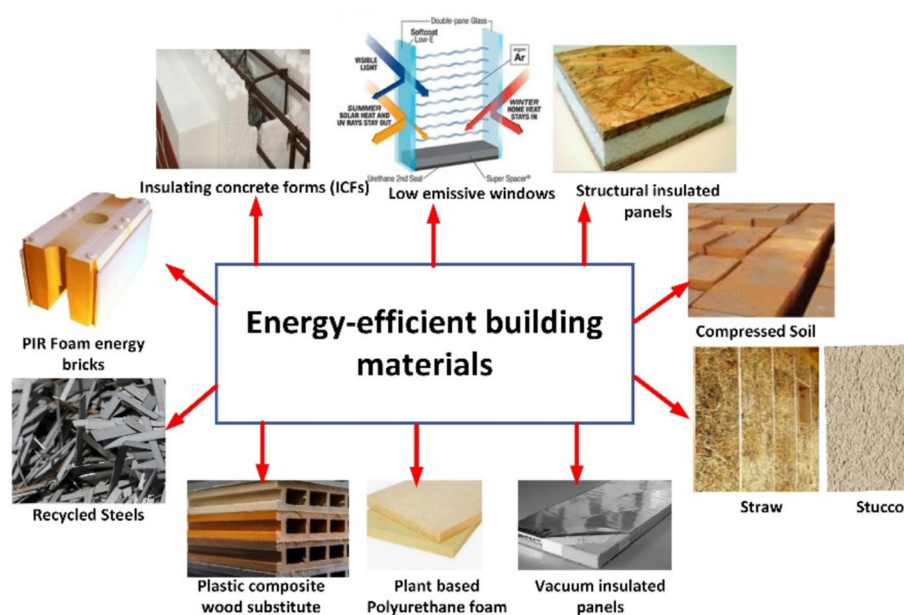


Fig. 3 Types of Energy-efficient building materials

Table 1 Energy-efficient building materials and their benefits

Material	Benefits	Limitations	Challenges
Insulating concrete forms (ICFs)	High energy efficiency, excellent soundproofing, strong and durable, resistance to natural disasters	Higher initial cost, requires specialized installation, limited design flexibility	Proper installation to avoid thermal bridging, moisture management, adapting traditional construction practices
PIR foam energy bricks	Excellent thermal insulation, lightweight, fire-resistant, moisture-resistant	Higher cost than traditional bricks, limited load-bearing capacity, potential off-gassing	Ensuring proper installation techniques, integration with other building systems, long-term durability concerns
Straw and stucco	Sustainable and eco-friendly, good insulation properties, low carbon footprint, affordable	Potential for moisture issues, fire risk if not properly treated, limited availability in some areas	Ensuring proper sealing against moisture, meeting building codes and regulations, changing perceptions about non-traditional materials
Compressed soil	Low environmental impact, excellent thermal mass, locally sourced in many areas, low cost	Labor-intensive construction, limited insulation properties, susceptible to erosion if not protected	Achieving consistent quality, meeting modern building standards, scaling up for larger construction projects
Structural insulated panels (SIPs)	High energy efficiency, quick installation, strong and lightweight, reduced thermal bridging	Higher upfront costs, limited design flexibility, potential for air quality issues	Ensuring proper sealing between panels, managing moisture and ventilation, training contractors in SIP construction
Vacuum insulated panels	Extremely high R-value, thin profile, lightweight	Very expensive, fragile and easily damaged, cannot be cut or modified on-site	Maintaining vacuum seal over time, integration with other building components, limited repairability
Plant-based polyurethane insulation	Renewable resource, low environmental impact, good insulation properties	Higher cost than traditional insulation, limited availability, potential for moisture absorption	Ensuring consistent quality, meeting fire safety standards, competing with established insulation materials
Plastic composite wood-substitute	Durable and low maintenance, resistance to rot and insects, recycled content options	Higher cost than natural wood, lower strength than natural wood, potential for thermal expansion	Improving structural properties, enhancing aesthetics to match natural wood, addressing end-of-life recycling
Low emissive windows	Improved energy efficiency, reduced heat gain/loss, UV protection	Higher cost than standard windows, potential for seal failure, may reduce beneficial heat gain in winter	Balancing solar heat gain and loss, improving durability of low-e coatings, educating consumers on benefits
Recycled steels	Reduced environmental impact, strong and durable, recyclable at end of life	Energy-intensive production process, potential for quality variations, higher cost than virgin steel in some cases	Ensuring consistent quality, improving sorting and processing techniques, increasing availability of high-quality recycled steel
Phase change materials (PCMs)	Thermal energy storage, reduces peak cooling/heating loads, enhances comfort	Higher initial cost, leakage risk, limited long-term stability	Optimal placement, integration into standard systems, improving cycling durability
Nanomaterials	Ultra-low thermal conductivity, self-cleaning, UV protection, air purification	High production cost, potential toxicity, lack of large-scale deployment	Regulatory frameworks, scalability, long-term health and environmental impact studies
Bio-based composites	Renewable, biodegradable, low embodied energy, good insulation/acoustics	Susceptibility to moisture, biodegradation, variable mechanical strength	Enhancing durability, fire resistance, quality control, standardization

5.1 Insulation materials

These materials are designed primarily to reduce heat transfer and improve the thermal envelope of buildings. They are widely used in walls, roofs, and floors to enhance energy efficiency.

5.1.1 Insulating concrete forms (ICFs)

Insulating Concrete Forms (ICFs) are a type of construction material used to build walls and foundations. ICFs consist of two panels made of rigid foam insulation, typically expanded polystyrene (EPS), that are held together by plastic ties or connectors [20–23]. These panels are designed to interlock, creating a hollow core into which concrete is poured, forming a solid, reinforced concrete wall. This type of construction provides excellent insulation, reducing the need for heating and cooling. ICFs offer several advantages over traditional construction methods. Because of their insulating properties, they can help reduce energy costs by keeping the interior of the building cooler in the summer and warmer in the winter [24]. They also provide superior sound insulation and are more resistant to fire, termites, and other pests than traditional wood-framed construction. Additionally, ICFs are relatively easy to install and can be used for a variety of building types, including residential, commercial, and industrial buildings. However, ICFs can be more expensive than traditional construction methods, and they require specialized training to install correctly [25]. They also require careful attention to detailing and waterproofing to prevent moisture from penetrating the walls. Despite these challenges, ICFs continue to gain popularity as a sustainable and energy-efficient building material.

5.1.2 PIR Foam energy bricks

PIR (polyisocyanurate) foam energy bricks, also known as PIR insulation blocks or PIR foam blocks, are a type of insulation material used in construction. PIR stands for polyisocyanurate, which is a type of rigid foam insulation material that is known for its high thermal resistance and excellent fire resistance [26]. PIR foam energy bricks are designed to be used as insulation for walls, roofs, and floors in residential and commercial buildings. They are lightweight and easy to handle, making them relatively simple to install. They are also highly efficient at reducing heat loss, which can lead to lower energy bills for building owners. One advantage of PIR foam energy bricks is their excellent fire resistance. PIR foam has a higher fire rating than many other insulation materials, making it a safer choice for buildings [27]. PIR foam energy bricks are also resistant to moisture, which can help prevent mold and mildew growth. However, there are some concerns about the environmental impact of PIR foam energy bricks. The production of PIR foam requires the use of chemicals that can be harmful to the environment, and the material is not biodegradable. Even, PIR foam energy bricks are more expensive than other insulation materials, although their high energy efficiency can help offset these costs over time.

5.1.3 Structural insulated panels (SIPs)

SIPs are made from a layer of insulation sandwiched between two panels of structural material. They are easy to install and provide excellent insulation [28–35]. These are a type of building material used for walls, roofs, and floors in residential and commercial buildings. SIPs are made from a layer of rigid foam insulation sandwiched between two layers of structural panels, such as oriented strand board (OSB) or plywood [36–42].

SIPs offer several advantages over traditional construction methods. They are highly energy-efficient, with a continuous layer of insulation that helps reduce air leaks and thermal bridging. This can lead to lower energy bills and a more comfortable indoor environment [43]. SIPs are also quick and easy to install, with prefabricated panels that

can be assembled on-site, reducing construction time and labor costs. Another advantage of SIPs is their strength and durability. The panels are highly resistant to wind, earthquakes, and other forces, making them a good choice for buildings in areas prone to natural disasters [44–46]. SIPs also provide excellent sound insulation, which can help reduce noise pollution in urban areas.

However, there are also some challenges associated with SIPs. The panels can be more expensive than traditional construction materials, and they require specialized training to install correctly. Additionally, SIPs can be susceptible to moisture damage if they are not properly designed and constructed, which can lead to mold growth and other issues [47, 48]. SIPs are a promising option for those looking for a highly energy-efficient and durable building material. With proper planning and construction, SIPs can provide a long-lasting and cost-effective solution for a variety of building types.

5.1.4 Vacuum insulated panels

Vacuum Insulated Panels (VIPs) are a type of building material used for insulation in walls, roofs, and floors of residential and commercial buildings. VIPs are made from a thin, gas-tight envelope that is evacuated of air, creating a vacuum [49]. The envelope is then encased in a protective barrier to create a panel. VIPs offer several advantages over traditional insulation materials. They have very high thermal resistance, or R-value, which means they can provide superior insulation with a much thinner profile than other materials [50–52]. VIPs are also very lightweight and can be easily cut to fit irregular spaces, making them a good choice for retrofit projects. Another advantage of VIPs is their durability. They are highly resistant to moisture, fire, and pests, and can provide excellent long-term performance [53]. VIPs can also be recycled, making them an environmentally friendly option. However, there are also some challenges associated with VIPs. They can be more expensive than traditional insulation materials, and they require specialized installation techniques to ensure that the vacuum is maintained [54–56]. VIPs can also be susceptible to punctures and other damage during installation or use, which can compromise their effectiveness. VIPs are a promising option for those looking for a high-performance and energy-efficient insulation material. With proper installation and maintenance, VIPs can provide a long-lasting and cost-effective solution for a variety of building types.

5.1.5 Plant-based polyurethane insulation

Plant-based polyurethane insulation is a type of building material made from renewable sources, such as soybeans, corn, or other plant-based materials [57]. Traditional polyurethane insulation is made from petroleum-based chemicals, which are non-renewable and have a significant environmental impact. Plant-based polyurethane insulation, on the other hand, is made from natural materials that are more sustainable and environmentally friendly. Plant-based polyurethane insulation offers several advantages over traditional insulation materials. It is highly energy-efficient, with a high R-value that can help reduce energy costs and improve indoor comfort. Plant-based polyurethane insulation is also non-toxic, with no harmful chemicals or VOCs that can impact indoor air quality [58]. Additionally, it can be easily installed in existing buildings or new construction, making it a versatile option for a range of applications. Another advantage of plant-based polyurethane insulation is its durability. It is highly resistant to moisture, fire, and

pests, and can provide excellent long-term performance. Plant-based polyurethane insulation can also be recycled or disposed of in an environmentally friendly manner, reducing its impact on the environment.

There are also some challenges associated with plant-based polyurethane insulation. It can be more expensive than traditional insulation materials, and it may not be readily available in all areas [59–61]. The manufacturing process for plant-based polyurethane insulation can be more complex than traditional insulation materials, which can impact its cost and availability. Plant-based polyurethane insulation is a promising option for those looking for a sustainable and environmentally friendly insulation material. With proper installation and maintenance, plant-based polyurethane insulation can provide a long-lasting and energy-efficient solution for a variety of building types.

5.2 Natural and recycled materials

These materials emphasize sustainability through the use of natural or repurposed resources, reducing environmental footprints.

5.2.1 *Straw and stucco*

Straw and stucco are natural building materials that have been used for centuries in various cultures around the world [62–64]. Straw is a renewable and widely available agricultural byproduct, while stucco is a traditional plaster made from cement, lime, and sand. In a straw and stucco wall system, straw bales are used as the primary structural element, and stucco is used to encase the bales and create a weatherproof barrier [65]. The bales are typically stacked like bricks, with the stucco applied to the exterior and interior surfaces of the wall. The resulting wall system is highly energy-efficient and provides excellent insulation. Straw and stucco building systems offer several advantages over conventional construction methods. They are sustainable and environmentally friendly, as straw is a renewable resource, and stucco is a natural material that can be made with minimal energy inputs [66]. Additionally, straw and stucco walls are highly resistant to fire, pests, and moisture. However, there are some challenges associated with using straw and stucco as building materials. The walls must be properly designed and constructed to ensure adequate structural support, and they may require specialized knowledge and skills to build [67, 68]. Additionally, straw and stucco walls can be susceptible to water damage if they are not properly protected, and they may not be suitable for all climates or building types. Overall, straw and stucco building systems are an excellent option for those looking for a sustainable and energy-efficient building material with a natural and rustic aesthetic [69–71].

5.2.2 *Compressed soil*

Compressed soil building material is a type of construction material made from a mixture of soil, cement, and water, which is then compressed into blocks or bricks using a hydraulic press [72]. This material is sometimes referred to as compressed earth blocks (CEBs) or stabilized compressed earth blocks (SCEBs). Compressed soil building material offers several advantages over traditional construction materials. It is a low-cost, sustainable, and environmentally friendly material that is readily available in many parts of the world. Compressed soil building material can also provide excellent insulation and thermal mass properties, helping to regulate indoor temperatures and reduce energy

costs [73]. Another advantage of compressed soil building material is its durability. When properly constructed and maintained, buildings made from compressed soil can last for many decades or even centuries. Additionally, compressed soil building material is fire-resistant and can provide good protection against earthquakes [74]. However, there are also some challenges associated with compressed soil building materials. The material can be heavy and difficult to transport, which may limit its use in certain areas. Compressed soil building material also requires careful design and construction to ensure adequate structural support, and it may not be suitable for all building types or climates. Compressed soil building material is a promising option for those looking for a sustainable, low-cost, and durable building material. With proper planning and construction, compressed soil buildings can provide a long-lasting and energy-efficient solution for a variety of building types.

5.2.3 Recycled steel

Using these energy-efficient building materials can help reduce energy consumption and costs, making buildings more sustainable and environmentally friendly [75–79]. Recycled steel is a building material made from scrap steel that has been processed and repurposed for use in construction. Steel is a durable and versatile material and recycling it can help reduce the environmental impact of construction projects [80–82]. Recycled steel can be used for a variety of building applications, including structural steel framing, roofing, siding, and reinforcement in concrete structures. Recycled steel has similar properties to new steel, including strength and durability, making it a reliable and long-lasting option for building projects.

Using recycled steel in construction offers several advantages over using new steel. First, recycling steel reduces the amount of waste that goes into landfills, which can help reduce environmental impacts [82]. Additionally, recycling steel requires less energy than producing new steel, which can help reduce greenhouse gas emissions and conserve natural resources. Finally, using recycled steel in construction can help reduce the overall cost of a project, as recycled steel is often less expensive than new steel [83]. However, there are also some challenges associated with using recycled steel in construction. Recycled steel may not be readily available in all areas, which can impact its cost and availability. Additionally, recycled steel may require additional processing to meet the specific requirements of a construction project, which can add to its cost [84–86]. Recycled steel is a promising option for those looking to reduce the environmental impact of their construction projects. With proper sourcing and processing, recycled steel can provide a durable and cost-effective solution for a range of building types.

5.2.4 Plastic composite wood-substitute

Plastic composite wood-substitute building material, also known as Wood Plastic Composite (WPC), is a type of building material made from a combination of wood fibers or flour and thermoplastic materials, such as polyethylene or polypropylene [87]. WPC is designed to mimic the look and feel of natural wood, while offering several advantages over traditional wood materials. WPC offers several advantages over traditional wood materials. It is highly durable and resistant to rot, decay, and insect damage, making it a good choice for outdoor applications such as decking, fencing, and cladding [88–90]. WPC is also low maintenance, requiring no sealing or staining like natural wood.

Additionally, WPC is more environmentally friendly than traditional wood materials, as it uses recycled materials and requires less energy to produce. Another advantage of WPC is its versatility [91]. It can be easily molded into a variety of shapes and sizes, making it a good choice for a range of applications. WPC is also available in a range of colors and finishes, allowing it to be customized to fit specific design needs.

But there are also some challenges associated with WPC. It can be more expensive than traditional wood materials, and it may not have the same aesthetic appeal as natural wood [92, 93]. WPC can be susceptible to fading or discoloration over time if it is exposed to direct sunlight. WPC is a promising option for those looking for a durable and environmentally friendly wood substitute. With proper installation and maintenance, WPC can provide a long-lasting and low-maintenance solution for a variety of building types.

5.3 Smart and adaptive materials

Smart materials adapt to environmental changes to enhance energy performance dynamically. Some of the smart materials and its benefits are enumerated below.

5.3.1 Phase change materials (PCMs)

Phase Change Materials (PCMs) are a class of smart materials that absorb and release thermal energy during phase transitions, typically from solid to liquid and vice versa [94]. This thermal regulation capability allows PCMs to maintain indoor temperatures within a comfortable range without relying heavily on mechanical heating or cooling systems [95–97]. When the ambient temperature rises, PCMs absorb excess heat and melt; when the temperature drops, they solidify and release the stored heat. This cycle contributes significantly to reducing peak thermal loads and improving energy efficiency in buildings. PCMs can be integrated into various building components such as walls, roofs, floors, and even glazing systems. Their ability to provide thermal inertia makes them ideal for use in climates with large temperature fluctuations.

Recent advancements have made PCMs more practical for construction, with materials like paraffin waxes, salt hydrates, and bio-based alternatives showing promise due to their high latent heat and appropriate melting points [98, 99]. Microencapsulation techniques have further enhanced their applicability by improving material stability and preventing leakage during phase change. Experimental and numerical studies have demonstrated that PCM-integrated systems can lower indoor temperature fluctuations by up to 6 °C and reduce cooling energy demand by more than 30% in hot climates. Despite these benefits, challenges remain, such as optimizing PCM placement, long-term stability, and cost-effectiveness [100]. However, as material science progresses, PCMs are expected to play a crucial role in the future of energy-efficient and climate-responsive building designs.

5.3.2 Nanomaterials

Nanomaterials are advanced materials engineered at the nanoscale (typically less than 100 nm) that exhibit unique physical, chemical, and mechanical properties not found in their bulk counterparts [101–103]. In the context of energy-efficient buildings, nanomaterials offer exceptional thermal insulation, self-cleaning surfaces, UV resistance, and air-purifying capabilities. Nanoparticles such as silica aerogels, carbon nanotubes, and

titanium dioxide (TiO₂) have been widely explored for applications in coatings, paints, concrete, and insulation systems. Due to their extremely low thermal conductivity, materials like silica aerogels can significantly reduce heat transfer, making them ideal for high-performance building envelopes. Moreover, the integration of nanomaterials into glass and window coatings enables selective solar control and visible light transmission, enhancing daylighting while minimizing unwanted heat gain.

Beyond insulation, nanomaterials contribute to improved indoor environmental quality. For instance, nanostructured photocatalytic coatings can break down organic pollutants and volatile organic compounds (VOCs), thereby purifying indoor air. Nano-enhanced concrete and composites exhibit superior strength-to-weight ratios, increased durability, and resistance to cracking and corrosion, which collectively contribute to lower maintenance and a longer building life cycle [104]. However, the widespread adoption of nanomaterials in construction faces certain barriers, including higher material costs, potential environmental and health concerns, and the need for regulatory frameworks. Continued research and real-world testing are essential to fully harness the potential of nanomaterials for creating energy-efficient, smart, and sustainable buildings.

5.3.3 Bio based composites

Bio-based composites are innovative building materials derived from renewable biological sources, such as plant fibers, agricultural residues, and biopolymers. These composites are designed to replace conventional petroleum-based materials while offering competitive or superior performance characteristics [105]. Common bio-based reinforcements include hemp, flax, jute, kenaf, and bamboo, which are combined with natural or synthetic binders like bio-resins or biodegradable plastics [106]. The resulting materials are lightweight, thermally insulating, and often biodegradable or recyclable, making them ideal for use in eco-friendly construction. Bio-based composites can be applied in a range of building components including insulation panels, wall boards, cladding, and interior partitions. Their low embodied energy and carbon footprint make them attractive for reducing the environmental impact of buildings across their life cycle.

In addition to their environmental advantages, bio-based composites often demonstrate excellent acoustic and thermal insulation properties, which contribute to improved occupant comfort and reduced energy consumption [107–109]. Some composites also have inherent resistance to mold, pests, and moisture, depending on the fiber treatment and binder type used. Despite these benefits, widespread implementation is still limited by issues such as variability in raw material quality, lack of standardized testing, and challenges in long-term durability under extreme environmental conditions [110, 111]. Ongoing research is focused on enhancing fire resistance, mechanical strength, and moisture stability of bio-based composites through hybridization and chemical modifications. As technological and regulatory support increases, bio-based composites are poised to become a vital component of sustainable and energy-efficient construction practices.

5.4 Fenestration and envelope materials

These materials improve the energy efficiency of a building's envelope by reducing unwanted heat transfer and solar gain.

5.4.1 Low emissive windows

Low-e windows are coated with a thin layer of material that reflects heat back into the building, reducing the need for heating. Low emissive (low-e) windows are a type of building material used for glazing in residential and commercial buildings [112–117]. Low-e windows are designed to reduce the amount of heat transfer through the glass, which can help improve energy efficiency and indoor comfort. Low-e windows are made by coating the glass with a thin, transparent layer of metal oxide [118–120]. This layer reflects radiant heat back into the room, preventing it from escaping through the glass. Low-e windows can also help prevent heat gain from the sun by reflecting some of the sun's energy back outside.

Low-e windows offer several advantages over traditional windows. They can help reduce energy costs by improving the insulation value of the windows. Low-e windows can also help improve indoor comfort by reducing drafts and temperature variations near windows. Additionally, low-e windows can help reduce the amount of UV radiation that enters a building, which can help prevent fading of interior furnishings and artwork [121]. However, there are also some challenges associated with low-e windows. They can be more expensive than traditional windows, which can impact the overall cost of a building project. Additionally, low-e windows can be more susceptible to scratches or damage during installation or cleaning, which can compromise their effectiveness. Low-e windows are a promising option for those looking to improve the energy efficiency and comfort of a building [122–124]. With proper installation and maintenance, low-e windows can provide a long-lasting and cost-effective solution for a range of building types [125].

Table 1 presents a comparative analysis of energy-efficient building materials discussed above [26–125]. Each material is analyzed not only for its direct thermal benefits but also its practical implementation challenges, such as cost, availability, and compatibility with existing building systems. The table aids in quick decision-making for material selection by highlighting a balanced view of advantages and disadvantages, and contextual considerations.

6 Sustainable construction components

Sustainability in building materials refers to the use of materials that have minimal impact on the environment and human health throughout their life cycle. This includes the extraction, manufacturing, transportation, installation, use, and disposal of building materials [126–129]. Sustainable building materials are those that are sourced and produced responsibly, with consideration for environmental and social factors [130]. They are durable, non-toxic, energy-efficient, and recyclable or biodegradable. Sustainable building materials help to reduce the environmental impact of construction by using resources more efficiently and producing less waste. This reduces greenhouse gas emissions, air and water pollution, and habitat destruction [131]. For example, recycled steel can reduce energy consumption by 75% compared to virgin steel production, significantly lowering the carbon footprint of construction projects. Using sustainable building materials can lead to long-term cost savings by reducing energy consumption, maintenance costs, and waste disposal fees. Additionally, some sustainable materials like timber can be less expensive than traditional materials like steel and concrete. Energy-efficient components, such as high-performance windows, can cut heating and cooling

costs by up to 30%, providing substantial savings over a building's lifetime. Sustainable building materials are often non-toxic and can help to reduce indoor air pollution, which can have a positive impact on human health. They also reduce the risk of exposure to harmful chemicals during manufacturing and disposal. For instance, low-VOC paints can decrease indoor air pollutants by up to 90%, creating healthier living and working environments.

Sustainable building materials can contribute to the well-being of communities by creating local jobs and supporting local businesses. They can also improve the aesthetics and comfort of buildings, which can have a positive impact on the quality of life for occupants. Using locally-sourced materials like timber can reduce transportation emissions by up to 50% while simultaneously boosting local economies. Many governments and municipalities require sustainable building practices and materials in new construction. By using sustainable building materials, developers and builders can ensure compliance with these regulations. Certifications like LEED now mandate the use of sustainable materials and practices, driving industry-wide adoption.

The use of sustainable building materials is essential for creating a more environmentally responsible and sustainable construction industry, which benefits the planet, people, and the economy. Life Cycle Assessments (LCA) play a crucial role in evaluating the true environmental impact of materials. For example, concrete with fly ash as a partial cement replacement can reduce CO₂ emissions by up to 20% over its lifecycle. Emerging trends in sustainable construction include bio-based materials like mycelium insulation and algae-based bioplastics, as well as smart materials that adapt to environmental conditions.

7 Performance assessment of building materials

Performance assessment of building materials can involve various criteria, including strength, durability, energy efficiency, safety, sustainability, and cost-effectiveness. Building materials such as concrete, steel, and masonry are typically evaluated for their compressive strength, tensile strength, and flexural strength [132–134]. Materials may also be subjected to cyclic loading or other stress tests to evaluate their durability and resistance to fatigue. Building materials can impact the energy efficiency of a building, with factors such as thermal conductivity and air leakage affecting a building's heating and cooling requirements [135]. Building materials may be tested for their insulating properties or evaluated through energy modeling software to determine their impact on a building's overall energy performance. Building materials must meet safety standards to ensure they do not pose a hazard to occupants of a building. Fire resistance, smoke toxicity, and other safety-related properties may be evaluated through standardized tests [136].

Sustainability is becoming an increasingly important consideration in building material selection. Materials may be evaluated based on their environmental impact, such as their carbon footprint, or their ability to be recycled or reused. The cost of building materials is a significant consideration in construction projects. Materials may be evaluated based on their cost-effectiveness, taking into account their durability, lifespan, and maintenance requirements [137, 138]. Building materials should be assessed based on the specific needs of a project and the context in which they will be used. A balance must be struck between performance criteria and cost-effectiveness to ensure that the building is both functional and economically viable.

7.1 Numerical evaluation

Bachir and Taieb [139] analytically and numerically evaluated integrating three passive measures such as phase-change substances, coatings with low emissivity, and material for thermal insulation into clay bricks that are hollow and often used for roofing to improve their thermal performance. They suggested that applying a coating with reduced emissivity to hollow surfaces reduced equivalent thermal conductivity by 55–70%. At an emissivity of 0.1, conductivity dropped to 0.027 W/m·K. Filling cavities with EPS insulation increased thermal performance, the greater the upgrading, the more important the filling ratio. Integrating PCM decreased internal heat flux was approximately 39% lower than bricks without PCM. The annual optimal PCM melt temperature was found to be 18 °C. Combining PCM and insulation in the same brick improved thermal resistance and inertia more than individually. Simulating bricks under outdoor conditions in Becher, Algeria showed bricks integrating low emissivity coating and PCM reduced inner heat flux by 82% and delayed external thermal wave by 4 h compared to traditional bricks. The optimum configuration was determined to be encapsulated PCM paired with low emissivity coating based on the hot, arid climate analyzed. Results provide a reference for designing hollow clay bricks with improved energy efficiency and thermal comfort for roofing in hot climates.

Afshoon et al. [140] experimentally evaluated the mechanical and microstructural characteristics of copper slag (CS) in self-compacting concrete (SCC) and steel fibers (SF). Increasing CS ratio improved SCC workability by up to 60% and compressive strength by up to 30%. Adding SF significantly improved SCC tensile strength (up to 60%) and fracture behavior through crack control and bridging. Optimal CS content increased from 20 to 40% with higher SF percentages (0.1% to 0.5%). SEM imaging showed CS refined pore structure and strengthened SCC matrix compared to plain SCC. Artificial neural network models more accurately predicted SCC strength characteristics (R^2 0.86–0.96) than multiple regression analysis. Using industrial CS byproducts reduces concrete production costs, energy use, and landfill waste; and enhances sustainability.

Alqahtani et al. [141] experimentally studied heat loss reduction in building materials using polymer-based lattice panels with varying porous geometries. Lattice panels were scaled up to 1 m² using a novel moulding technique maintaining U-values despite the unit cell size. An experimental scaling law accurately predicted lattice panel thermal performance. Hydraulic diameters under 8 mm showed only conductive heat transfer. Higher temperature gradients across lattice panels improved U-values and thermal resistance. Optimal sensor location and testing recommendations proposed for optimal lattice performance. 30 °C temperature exposure tests in the calorimeter box and guard hot box showed significant heat loss reduction. Commercial viability is demonstrated with easy production and customizable geometries. Polymer lattice panels are promising candidates for effective thermal insulation for the use of building envelopes. They can contribute to increased residential energy efficiency and reduced heating/cooling loads.

Al-Yasiri and Szabó [142] performed a numerical analysis of thin building envelope-integrated phase change material towards energy-efficient buildings in severe hot locations. They numerically evaluated the phase change material (PCM) based on paraffin's thermal performance with a melting temperature of 40–44 °C integrated passively into the roof and walls of a small room model under severe summer conditions in Al Amarah, Iraq. PCM integration reduced average indoor temperature fluctuation by 5–6 °C

and the thermal load leveling by 38–59% compared to a room without PCM. The PCM room showed an average operative temperature reduction of 6 °C, improving thermal comfort despite high ambient temperatures. The PCM room had 66.6–76.5% lower average heat gain than the reference room, with the roof providing the largest contribution. Equivalent CO₂ emission savings of ~2 kg per day and electricity cost savings of 210–250 IQD per day were achieved with PCM integration. PCM thermal performance relied strongly on diurnal ambient temperature cycles indicating the need for night ventilation or insulation to improve solidification. PCM incorporation displayed good potential to passively enhance thermal comfort and reduce cooling demands in extremely hot climates when properly designed.

A numerical model was developed to investigate PCM-integrated bricks' thermal performance under different indoor conditions, wall orientations, PCM positions inside the brick, and PCM thicknesses [143]. The model was validated with good agreement against experimental data from the literature. Simulations were carried out for the climate of Jaipur, India on the hottest day of the year. Different PCM materials like paraffin wax and fatty acids were analyzed. It was found that PCM integration significantly decreased the amount of heat entering the room for all materials, orientations, PCM positions, and thicknesses studied. The maximum heat flux reduction achieved was over 60% compared to a brick without PCM. Optimization studies revealed that a PCM thickness of 30–40 mm placed near the outdoor side of the brick results in the best thermal performance in terms of lowering indoor heat gain. Higher reductions were achieved for east and west-facing walls compared to north/south walls. Unconditioned indoor spaces saw much more heat flux reduction than air-conditioned spaces. An economic feasibility study found attractive payback periods of around 3 years for PCM integration in the Jaipur climate considering savings from space cooling equipment downsizing.

A mathematical model was created and validated in the study where the thermal performance of PCM coatings was applied to building wall structures [144]. Simulations were carried out for 8 different wall types and 4 European climates. It was found that PCM coatings can significantly reduce the heat flux through the walls for all cases, with reductions of up to 52% compared to the walls without PCM coatings. The optimal phase change temperature range for PCMs was found to be 18–26 °C for the European climates studied. PCM coatings with phase change in this range provided the best thermal regulation. The study analyzed the sustainability impact using a life cycle analysis approach. PCM coatings increased the global warming potential compared to conventional coatings but reduced other impact categories. An energy payback period analysis showed that the added embodied energy of PCM coatings can be recovered within 1–8 years through savings in the building heating/cooling loads. The optimal PCM coating thickness was determined to be 3–5 mm beyond which heat flux reductions marginally improved. Thicker coatings also increase costs and embodied energy.

Elnajjar [145] developed a numerical model to study the energy savings from integrating different PCM materials into building bricks commonly used in UAE buildings. Simulations were carried out for one day and seven days. It was found that the choice of an optimal PCM is not straightforward and can be deceiving if only 1-day performance is evaluated rather than longer duration. The n-Octadecane PCM performed best on the first day showing significant temperature reduction. However, over 7 days the RT25 PCM showed better sustained performance. Additionally, the PCM's placement inside

the brick impacted performance. Placing the PCM towards the indoor side was better than placing it near outdoor. Quantitatively, the use of an optimized PCM-brick configuration resulted in a peak temperature reduction of 3 °C and up to 35% savings in daily cooling loads compared to brick without PCM. The results highlighted that rigorous simulation-based analysis over sufficiently long periods spanning multiple PCM melt/freeze cycles is essential to identify the most appropriate PCM option for specific climates and building materials.

Borreguero et al. [146] developed an equation for heat conduction based on Fourier analysis to predict the thermal behavior of gypsum wallboards incorporated with different percentages of microencapsulated phase change materials (PCMs). The model was validated experimentally using a custom test setup. Good agreement was found between model predictions and experiments for gypsum samples with 0%, 4.6%, and 6.4% microencapsulated PCMs by weight. It was shown that by using a temperature-dependent apparent heat capacity obtained from differential scanning calorimetry, the latent heat effects of PCMs could be captured without needing to explicitly track the solid–liquid interface. Increasing the PCM percentage enhanced the heat storage capacity of the wallboards, with around 12–17% improvement for the PCM percentages tested. The study found that using 5% microencapsulated PCM by weight allows reducing the thickness of the gypsum by 8.5% while keeping the same degree of thermal comfort. The validated model provides an effective tool to predict thermal performance improvements from integrating microencapsulated PCMs into gypsum drywalls and can help optimize building designs accordingly.

Shahrzad and Umberto [147] performed parametric optimization of multifunctional integrated climate-responsive opaque and ventilated façades using CFD simulations. They used parametric models to enhance the MICRO-V façade's design, a novel climate-responsive structure. MICRO-V is a multifunctional, integrated, climate-responsive, opaque, and ventilated façade that was created to enhance indoor environmental quality. They Controlled the passage of heat, air, and moisture between indoor and outdoor environments in response to climate variations. They used integrated ventilation modules to pre-condition supply air and enable heat and moisture recovery. They evaluated the effects of various design elements on the functionality of Micro-V and demonstrated using façade-scale models for decision-making and prototype development. Increasing the ventilation module's heat transfer area improves pre-conditioning efficiency.

As mentioned in the literature the growing impacts of climate change have created the urgent need to adopt sustainable technologies in the building sector [148]. Static thermal insulation alone cannot handle rising energy demands and indoor comfort issues.

Amanowicz et al. [149] highlighted the advances in energy-efficient ventilation systems for buildings over the past 3 years. Building envelope airtightness is a prerequisite for efficient ventilation system performance. Demand-controlled ventilation (DCV) reduces ventilation airflow and associated fan energy use. Decentralized systems avoid duct losses associated with central systems. Mechanical ventilation with heat recovery is essential for indoor air quality and thermal performance regardless of building type. Main ventilation system requirements are airtight construction, controlled/optimized airflow, ductless or short ducts, and exhaust air heat recovery. The review reinforces the need for revised policies, decentralized systems, ventilation control methods, and heat recovery to improve building energy efficiency. It identifies research gaps and technology opportunities in ventilation system design.

7.2 Experimental evaluation

In an experiment conducted by Al-Yasiri and Szabó [142] over 7 days in August–September 2020 in hot climatic conditions in Al Amarah, Iraq, 4 identical test rooms with different roof configurations evaluated: Room A (control), Room B (PCM layer between finishing and main concrete layer), Room C (PCM layer in the middle of concrete layer), Room D (PCM layer between concrete and cladding layer). PCM used was paraffin wax with a melting temp of 44 °C. Model B showed the best performance in Room Max Temp Reduction (RMTR) with a max 12.9% reduction. Model C showed the best performance in Average Temperature Fluctuation Reduction (ATFR) with a 7.1–9.5 °C reduction. Model B had the lowest Decrement Factor (DF) indicating best utilization of PCM to moderate inside temp fluctuations. Model C is the second best.

Experimental Models Tested by Hasan et al. [150] where rooms with all walls insulated with 1 cm thick PCM, Room with only ceiling insulated with 1 cm thick PCM, Rooms with individual walls (west, south, north, east) insulated with 2 cm thick PCM. Indoor Temperature Reduction for 1 cm PCM on all walls was 2.18 °C. 1 cm PCM on the ceiling only reduces the temperature by 1.9 °C. 2 cm on the south wall gave a max reduction of 3.32 °C. 1 cm PCM on all walls gave a 20.9% peak reduction. 2 cm on the south wall gave a 19.95% peak reduction. Ceiling PCM gave a minimum 6.83% reduction. PCM showed a good potential for cooling load savings in buildings.

Researchers analyzed and compared the indoor thermal behavior of two experimental cubicles—a reference cubicle and one with macro-encapsulated PCMs integrated into the walls to regulate temperature [151]. Data was collected over 24 h. Temperatures were recorded for the south, west, north, and east walls, roof, and indoor ambient air of both cubicles. During the day, the PCM cubicle's walls, roof, and interior air all displayed lower peak temperatures than the reference cubicle, demonstrating the PCMs' ability to store thermal energy and prevent overheating. At night, the PCM cubicle temperatures remained warmer than the reference as the PCMs released their stored latent heat. This shows the need for ventilation to dissipate this heat. The PCM cubicle showed 40–60% lower thermal amplitude (range between daily max and min temps) for all surfaces, indicating increased temperature stability. Time lags of 60–120 min were achieved for the different cubicle elements before reaching peak temps, demonstrating the potential for cooling load shifting. Integration of macro encapsulated PCMs into building envelope elements can regulate temperatures, reduce amplitude, improve thermal stability, and delay peak temps. This can decrease dependency on mechanical cooling and shift/reduce peak loads, saving energy. Proper ventilation is needed to dissipate stored thermal energy released by PCMs at night. Heat flux sensors on the walls and roof measured heat transfer rates in the reference and PCM cubicles. Before peak temps, PCM cubicle heat fluxes were lower as PCMs absorbed heat during melting. After peaks, they were similar or slightly higher as PCMs released heat. The roof showed the minimum heat transfer reduction (19.41%), while the east wall showed a maximum 41.31% reduction due to the PCM integration. For the PCM cubicle, the overall peak heat flow was 27.32% lower. Infrared thermography confirmed lower average wall temperatures for the PCM cubicle at different times of the day. This supports the temperature measurements. Microencapsulated PCM integration substantially reduced heat transfer into the cubicle, cutting cooling loads by over a third. This translated into quantifiable electricity cost savings for space cooling. Thermographic imaging visually verified and complemented the temperature data, showing PCM walls had lower temperatures.

Experimental analysis conducted by Zhao et al. [152] stated that Infrared imaging can show the phase change material (PCM) in its full liquid state, with the metal foam structure visible. Temperature profiles showed periodic variations due to differences in PCM and metal foam structures. Adding metal foam allows heat transfer through conduction even when the PCM is solid, leading to more uniform temperatures. During melting, the metal foam sample wall and PCM temperature difference were much lower than for pure PCM, showing 5–20 × heat transfer enhancement. This difference decreased slightly during phase change and then sharply after full melting due to natural convection effects. Smaller metal foam pore size and higher density further improved heat transfer rates. Metal foams improved the heat extraction rates during cooling/solidification as well. Good agreement was achieved between experimental data and numerical simulations.

Korjenic et al. [153] developed and tested materials for thermal insulation made of hemp, flax, and jute fibers, among other renewable agricultural resources. They assessed the materials' building physics properties and performance compared to conventional insulations. They examined the impact of moisture on properties over time. They created 6 sets of test specimens from varying compositions of natural fibers, shives, and binders. They tested properties like density, thermal conductivity, vapor diffusion, moisture absorption, etc. per building material standards. The effect of increasing moisture was evaluated on the thermal conductivity. They simulated performance using measured properties. Natural insulation specimens showed thermal conductivity from 0.039–0.049 W/mK, on par with conventional materials. Mechanical properties like compression strength and tensile strength are also adequate.

Vollaro et al. [154] compared energy simulations from steady-state (MC11300) and dynamic (TRNSYS) building energy software. They evaluated the effect of different window glazing options on heating/cooling demands and assessed the thermal behavior and inertia of the building envelope. They performed simulations using MC11300 and TRNSYS on a case study building, compared heating/cooling demands for models with single, double, and double argon-filled glazing, and analyzed temperature profiles and thermal inertia when the HVAC system is on/off. MC11300 results diverged from TRNSYS by 12–14%, underestimating heating and overestimating cooling. The difference is likely due to overestimated solar gains and lack of thermal mass decrement factors in MC11300. Improved glazing reduced heating demands in both models, but increased cooling demand in the steady-state model, Dynamic TRNSYS results showed smaller cooling demands.

Liu et al. [155] revealed that MC11300 (steady-state simulation) consistently underpredicts heating demands and overpredicts cooling demands compared to TRNSYS (dynamic simulation)—by about 12–14% in this case study. The difference is likely due to MC11300's simplified solar gain model not accounting for thermal mass effects, which are important for buildings like historical structures with massive walls. The discrepancy between steady-state and dynamic simulation is expected to increase for buildings with higher wall thicknesses and thermal mass. For insulating windows, MC11300 predicts cooling demand rises while heating drops since transmission losses reduce faster than solar gains.

Berardi and Soudian [156] introduced BeTOP, a new complete outdoor test capability located in Toronto, Canada for experimental performance evaluation of advanced building materials, components, and systems. BeTOP had two identical test rooms allowing

comparative testing of up to 8 full-scale façade systems and 2 ceiling systems simultaneously under real outdoor conditions. The rooms also allowed flexible control of indoor conditions. The test cell was designed to study hygrothermal performance, energy efficiency, indoor environmental quality, and integration of renewable energy systems into buildings. It enables testing from material to whole-building scale. BeTOP filled a research gap in Canada for testing systems under continental climate conditions with cold winters and hot, humid summers.

Gracia Cuesta et al. [157] have tested different passive insulation systems (polyurethane, mineral wool, polystyrene) in building envelopes to evaluate their performance for heating and cooling. Insulated test cubicles showed smaller temperature fluctuations (0.5–1 °C) compared to non-insulated ones (1.5–2 °C) under free-floating conditions. Polyurethane insulation provided the greatest energy savings, which were around 36% and 58% compared to other materials during summer and winter. After 7 years, there was no noticeable degradation in insulation performance. Energy savings of around 34–60% were achieved during cooling.

A comprehensive framework was suggested for evaluating and characterizing the performance of adaptive façade systems using numerical simulation and experimental testing methods [158]. Standardized quantitative approaches for assessing different aspects of adaptive façade performance were established as building technologies continued advancing rapidly.

An innovative test cell has been designed for evaluating building façade components under real Mediterranean climate conditions [159]. A novel heat flux sensor tile was developed with 5 times higher sensitivity compared to prior sensors. The new sensors measured heat flux over a 530 × 530 mm area. Unlike previous small-area sensors, the new tile provided uniform flux over the whole area, enabling reliable characterization of full-scale components. The test facility and sensing methods aimed to enable realistic quantification of dynamic thermal performance factors for innovative building façade technologies and materials.

A design optimization process was suggested for an innovative test cell to evaluate building facade technologies under real climate conditions in Central Europe [160]. The test cell was built from a modified shipping container structure and contained removable facade segment samples mounted on two opposite walls. Numerical simulations were conducted using finite element and finite volume methods to optimize the test cell and segment box configurations for accurate thermal measurements. The test segments were enclosed in small test boxes to isolate them and reduce temperature fluctuations. The boxes were constructed to be approximately 1 cubic meter in volume. Simulations modeled annual transient heat transfer for different facade transmittance and resistance properties. Results showed solar transmittance has an approximately linear relationship with annual heat flow. The final test cell design incorporates controlled air conditioning, extensive sensor arrays in each box and test segment, and automation for data collection over long durations in real climate conditions.

As highlighted by Claros-Marfil et al. [161], adaptive water-flow glazing's (WFGs) have the potential to enhance building energy efficiency, but their dynamic properties are difficult to simulate in standard building energy modeling software. A new experimental simulation methodology using TRNSYS was presented to ascertain the solar heat gain coefficient (SHGC) and U-value of WFGs for mild climates.

Saber et al. [162] carried out numerical calculations to assess the thermal efficiency of residential wood-frame walls retrofitted with vacuum insulation panels (VIPs) or XPS foam. Results showed VIP walls provided much higher R-values (up to 55 ft²h°F/BTU) compared to XPS foam retrofits (R-29.6 ft²h°F/BTU). Thermal bridging through the VIP envelope can significantly reduce its effective R-value, but this effect was not quantified here. Transient simulations were validated with experimental heat flux measurements showing good agreement. Simulations predicted 69–79% lower annual heating loads for VIP walls compared to the XPS foam wall. Adding a furred airspace on the interior of the VIP wall provided a further 5% heating load reduction compared to VIP alone.

A novel insulation panel concept using vacuum insulation panels (VIPs) is proposed for anchoring green wall systems to building facades [163]. The panel combines VIPs with extruded polystyrene (XPS) and polyurethane (PUR) foam to achieve high thermal performance in a thin, formattable structure. Grids of VIPs separated by PUR gaps allow the panel to be shaped and anchored while minimizing thermal bridging. Numerical simulations optimized the VIP size, thickness, and gap width for thermal resistance and anchoring functionality. A 30 mm thick VIP with 50 mm PUR gaps provided a 220 mm EPS-equivalent performance in a 70 mm panel, reducing insulation thickness by 67%. The model showed a maximum 1.4 K temperature drop at VIP-PUR transitions and negligibly affected overall thermal resistance. The thin, structurally robust panel enables simplified integration of green wall systems on building facades.

Over the course of five years, an innovative vacuum insulation panel (VIP) composite integrated with polyiso roof insulation was assessed [164]. In winter conditions, the VIP composite demonstrated a 40% reduction in heat flux when compared to the polyiso insulation alone. According to laboratory studies, the VIP composite performs better at low temperatures than polyiso alone, showing greater thermal resistance. In both winter and summer, the VIP composite outperformed the polyiso system in insulation, resulting in deck temperatures that were 4–5 °C lower.

Comparative experiments were conducted to assess the energy and thermal efficiency of opaque ventilated facades (OVFs) in China's cold temperature zone using two complete test apparatuses [165]. Limitations on cavity ventilation dominated the winter routine, while characteristics controlling surface solar radiation had the most impact on summer performance. The suggested OVF layouts for the summer and winter resulted in energy savings of 11.4% and 6.5%, respectively, compared to conventional facades. OVFs are appropriate for daytime-use buildings since they showed greater daytime energy savings. Using normalized analysis, an integrated OVF design was suggested for best annual performance and least amount of complexity. The results include a representative sample for China's cold climatic zone as well as basic guidance for OVF design.

Ciampi et al. [166] carried out comparative studies on two large test rigs to evaluate the energy and thermal efficiency of opaque ventilated facades (OVFs) in China's cold environment. The thermal behaviors were greatly affected by OVF design elements such as cavity width, air vent openness, outer cladding color, and joint opening ratio. Limitations on cavity ventilation dominated the winter routine, while characteristics controlling surface solar radiation had the most impact on summer performance. The suggested OVF layouts for the summer and winter resulted in energy savings of 11.4% and 6.5%, respectively, compared to conventional facades. OVFs are appropriate for daytime-use buildings since they demonstrate greater daytime energy savings. By using a normalized

analysis of the evaluated façade, an integrated OVF configuration was suggested for the best annual performance with the least amount of complexity. Results provide general OVF design guidelines and a climatically representative case study for China's chilly temperature zone.

Designing energy-efficient buildings and lowering energy usage depends on the choice of energy-efficient building materials [167]. Building materials are judged on their energy efficiency based on several factors, such as their locality, ability to be recycled, use of natural renewable resources, labor-intensiveness, needing less energy-intensive manufacturing processes, use of renewable energy in production, low thermal conductivity, and reduced energy consumption during construction. Wood, pumice, perlite, gypsum mud bricks, cellular concrete, cellulosic insulation with boron, aerogels, and ceramic foam insulation were among the common building materials that were examined.

A conceptual framework was presented to support integrated thinking on how building materials can contribute to achieving the UN Sustainable Development Goals (SDGs) [168]. A 7-point scale adapted from previous research is used to score the interaction of different building materials typologies with various SDGs. The framework revealed building materials significantly influence 13 SDGs and 25 targets related to wellbeing, energy/water efficiency, sustainable production/consumption, resilient infrastructure, climate change mitigation etc. Green, alternative, eco-friendly and responsibly sourced building materials showed the most potential to realize multiple SDGs.

Vanadium dioxide (VO₂) films [169] were deposited by RF sputtering on fused silica and commercial float glass substrates at 300 °C. Films on fused silica showed low crystallinity due to the low deposition temperature. Thinner films (30 nm) on fused silica exhibited a very low critical transition temperature (T_c) around 33–37 °C, attributed to defects from low crystallinity and non-stoichiometry. However, they had poor IR switching and solar modulation.

Marble dust waste from three different regions in Pakistan (Mohmand Agency, Buneer, and a mix of Ziarat-Mullagori) was used to partially replace clay in fired bricks up to 30% by weight [170]. Adding marble dust decreased the bulk density, compressive strength, and thermal conductivity of the bricks, while increasing the porosity and water absorption. Up to 20% marble dust addition satisfied the minimum compressive strength requirement of 7 MPa for individual bricks as per standards. The maximum reduction in thermal conductivity was 13.35% at 30% marble dust addition compared to control bricks, indicating improved insulation properties. The U-values (heat transfer rate) of the bricks decreased with increasing marble dust content due to higher porosity. No efflorescence issues were observed in any of the marble dust-added bricks. The properties did not vary significantly between the three marble dust sources due to their similar chemical compositions. Utilizing marble dust waste in fired clay bricks was found to be a viable waste management solution that can produce economical and environmentally friendly building materials.

7.3 Analytical evaluation

Researchers suggested the BEPAS (Building Environmental Performance Assessment Score life cycle assessment (LCA) model to gauge the impact on the environment of buildings over the course of their 50-year life cycle [170]. The effects of building operation (water and energy use), building material production and transportation, and building surrounds (landscaping, etc.) were considered in the model. It adhered to the four

LCA steps: inventory analysis, effect analysis, goal/scope definition, (classification, characterization, valuation), and interpretation. For valuation, a novel weighting method was suggested to use green taxes based on social willingness-to-pay (WTP) to combine many environmental effects into a single eco-point score.

A comprehensive review of recent applications of artificial intelligence (AI) to structural design and performance assessment (SDPA) issues in buildings was provided by researchers [171]. The ML-SDPA studied fell into four main categories: (1) predicting the response and behavior of structures; (2) modeling using real experiment data; (3) retrieving information from text and images; and (4) modeling using data from structural health monitoring and field reconnaissance. For each category, a list of the machine learning techniques was provided. The paper identified several key challenges facing the wider adoption of ML for SDPA problems, including data scarcity and quality issues, lack of consensus on algorithm selection, poor explainability and interpretability of ML models, and overfitting concerns.

A Mathematical model was developed by Akanbi et al. [172] to calculate the lifetime of building materials, using data on the usual life expectancy, reusability, and end-of-life performance of building materials. The model was evaluated employing a case study of a two-story residential structure, including the material take-off measures for components like foundations, floors, frames, stairs, and walls. The results show that as a building ages, the quality and reusability of recoverable materials diminishes. Structural components like concrete retain some quality beyond 100 years, while non-structural components like softwood and plasterboard degrade much faster. With the model, designers may simulate material performance over its complete life and adjust for more effective material recovery that adheres to the circular economy's guiding principles.

A novel approach to assess the potential for improving energy consumption and indoor climate in existing industrial facilities was studied through dynamic thermal simulation modelling [173]. Two refurbishment scenarios were proposed and evaluated for an existing metal processing factory i.e. one insulating the existing roof and keeping skylights (S1), and the other replacing the roof with new insulated panels and redesigned skylights (S2). Simulation results showed S1 reduced total facility energy demand by 16% and S2 by 23% compared to the base case. The significantly better thermal performance of the nano paints over ordinary paints was demonstrated by the reduction of up to 40% in the rates of heat exchange via the outer envelope for both the internal and external surfaces, and up to 30% for the external surfaces alone [174]. The heat transfer values considerably increased when nano-insulating layers were utilized in the external walls rather than more traditional materials like polystyrene foam. Because of the envelope's eightfold lower thermal transmittance coefficient (U-value) than that of traditional insulation, the heat transfer rates were lowered by 45%.

Researchers present a brand-new mixed-integer nonlinear programming (MINLP) paradigm that is multi-objective to optimize material selection for building design to minimize heat gain, maximize daylighting, and minimize net present cost [175]. The model was applied to two case studies—a residential flat and a commercial building. 33% improvement in daylighting, 17% cost reduction, and 30% less heat gain were noted for residential flats. For the commercial 43% improvement in daylighting, 23% cost reduction, and 39% less heat gain were marked. The case studies highlighted the potential for significant improvements in human decision-making by using the optimization model.

The most recent developments in thermal and acoustical insulating materials made of recycled or natural materials were reviewed by many researchers [176]. Insulating materials were categorized based on heat exchange properties, shape (loose-fill, rigid boards, spray foam, batts/blankets), and content (inorganic, organic, mixed materials, materials utilizing modern technology). A detailed study of phase change materials (PCMs), vacuum insulation panels, closed cell foams, gas-filled panels, and aerogels was done. The functionality of a real-world active phase change material (PCM) system was evaluated over 25 months using monitored data [177]. The PCM system contributed 12–37% of the whole cooling capacity during winter months but remained inactive during summers due to insufficient night-time temperatures for charging. The actual energy savings were much lower than the 33% predicted savings, except for a few winter months. Key reasons for the performance gap included a lack of chiller integration for PCM charging, differences between actual and specified PCM properties, control logic issues, and a lack of training for operators. The study highlighted the importance of evaluating active PCM systems in real buildings to identify and address performance gaps from the design stage.

While building energy policies have had some success in reducing energy intensity, significant barriers to wider implementation remain, including high upfront costs, lack of financing, institutional issues, behavioral factors, and lack of information/awareness [178]. Overcoming these barriers will require coordinated efforts involving regulations, financial incentives, technological development, information dissemination, and addressing societal norms. Buildings account for a significant portion of worldwide power use, around 40% in the U.S. and EU, and 27.3% in China. Reducing building energy consumption is crucial for mitigating climate change impacts. A three-step approach was suggested for achieving net zero energy buildings (ZEBs): passive energy savings, energy-efficient building systems, and renewable energy generation [179]. Technologies like improved envelopes, daylighting, HVAC optimization, and solar/wind/geothermal were reviewed. Climate By 2100, changes were predicted to result in a 34% reduction in the world's heating demand but a 72% increase in cooling demand, with larger increases in cooling needs for developing warm regions [180].

A multi-objective optimization methodology was proposed using an algorithm to minimize both life cycle cost and CO₂ emissions for residential building design [181]. When evaluating costs and emissions, it considered all phases of a building's life cycle: pre-use (material extraction, manufacturing, transportation), use phase (operational energy, maintenance), and end-of-life (demolition, waste transportation). The results showed that emission reduction can only be achieved by increasing life cycle costs. The optimization approach provides valuable insights for architects/designers to assess the environmental impact vs economic costs for different building envelope designs early in the design process.

A new approach was suggested that integrates life cycle cost analysis (LCC), life cycle assessment (LCA), and multicriteria decision analysis (MCDA) to comprehensively evaluate alternative design solutions for energy-efficient buildings [182]. Results showed that for an energy-efficient home, the energy that is embodied from construction can account for over 1/3 of the total primary energy consumption during the life cycle, much higher than for standard buildings. This emphasizes the importance of carefully selecting construction materials and systems. Comparing masonry, and timber frame construction

alternatives, the log building performed best overall across environmental impact categories like energy demand and emissions, though it had slightly higher life cycle costs. Energy-efficient design strategies were suggested across all phases of the lifespan of a structure—prebuilding, building, and post-building phases [183]. For the pre-building phase, important factors like site selection, building orientation, form, envelope, material choice, landscaping, and use of renewable energy sources through passive and active techniques were discussed.

The pre-building phase is crucial for achieving energy efficiency in buildings. Careful site selection that considers factors like latitude, slope, prevailing winds, and sun exposure can significantly impact a building's heating, cooling, and ventilation needs [184]. The building's orientation, form, and envelope design play a vital role in maximizing useful solar gains while minimizing unwanted heat losses or gains. The choice of building materials with good insulating properties, low embodied energy, and locally sourced origins reduces energy consumption during construction and operation.

According to the framework described by researchers, the building envelopes can be made more environmentally friendly and energy-efficient by integrating Building information modeling (BIM), life cycle assessment (LCA), and mathematical optimization methods [185]. To determine the best combination of building envelope materials, uncontrollable factors were identified like temperature, humidity, and airflow to achieve an energy-efficient building design for a typical Malaysian residential house [186]. It used a novel approach combining building information modeling (BIM) simulations with the statistical Taguchi method to evaluate and optimize building component materials while accounting for uncontrollable environmental factors. BIM simulations identified the most energy-efficient material options for floors, walls, windows, ceilings, and roofs based on cooling load reductions. The Taguchi analysis ranked the ceiling material as having the greatest influence on energy optimization, followed by windows, walls, roofs, and floors in descending order. The combined BIM-Taguchi approach provided insights into optimizing building component selection by accounting for both controllable material properties and uncontrollable environmental conditions. BIM enables comparative analysis of design options like building orientation, envelope materials, and window sizing to optimize energy performance. While BIM supports energy estimation, there are opportunities to incorporate real-time consumption monitoring, occupant behavior impacts, and country-specific energy standards [187, 188].

8 Conclusions and recommendations

This review systematically analyzed a broad spectrum of energy-efficient building materials categorized into insulation systems, natural and recycled alternatives, and smart adaptive materials. Smart materials like Phase Change Materials (PCMs), nanomaterials, and bio-based composites demonstrated significant potential for reducing energy consumption and enhancing indoor thermal comfort. Meanwhile, natural and recycled materials such as straw, compressed soil, and recycled steel support sustainability goals through low embodied energy and resource reuse, though their adoption is often limited by performance variability and regulatory challenges. Traditional and advanced insulation materials continue to form the foundation of energy-efficient construction, providing reliable thermal resistance and environmental benefits despite higher initial costs.

The study further highlighted the importance of experimental, numerical, and analytical evaluations in validating material performance. For broader implementation, challenges such as cost, integration complexity, standardization, and awareness must be addressed. Moving forward, an integrated approach combining material innovation, policy support, and interdisciplinary collaboration is essential. These findings provide valuable guidance for designers, researchers, and policymakers committed to building climate-resilient, low-carbon structures that align with global sustainability objectives and the UN Sustainable Development Goals.

Certain recommendations are proposed to enhance the adoption and effectiveness of energy-efficient building materials in sustainable construction practices. Performance benchmarks and certification protocols should be established for emerging materials such as phase change materials (PCMs), nanomaterials, and bio-based composites to ensure quality, safety, and comparability. Supportive policies, subsidies, and tax incentives should be implemented to reduce the high upfront cost of advanced energy-efficient materials and promote their adoption in public and private construction projects. Collaboration among researchers, policymakers, material scientists, and construction professionals should be promoted to align innovation with regulatory frameworks, construction practices, and sustainability goals.

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